

MATH 301: Homework 1

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Question 1: From Blackboard

Proposition: Let S and T be two sets. If both S and T are countable, then $S \cup T$ is countable.

Note: If S and T are said to be infinite and countable; then they are said to be *denumerable*

Proof. Suppose S and T are countable, and there exists $A, B \subseteq \mathbb{N}$, by definition there exists a bijective function:

$$f : A \rightarrow S \text{ and } g : B \rightarrow T$$

To show $S \cup T$ is countable, we must construct a bijection $h : C \rightarrow S \cup T, C \subseteq \mathbb{N}$

Define $h : C \rightarrow S \cup T$:

$$h(n) = \begin{cases} f\left(\frac{n}{2}\right) & \text{if } n \text{ is even} \\ g\left(\frac{n+1}{2}\right) & \text{if } n \text{ is odd} \end{cases}$$

Note S, T must be disjoint; else $\exists n_1, n_2 \in C, n_1 \neq n_2 \ni h(n_1) = f(n_1) = h(n_2) = g(n_2)$

Thus, for the sake of the proof, presume S, T to be disjoint sets s.t. $S \cap T = \emptyset$

To prove the proposition we must show that h is *surjective* and *injective*.

Surjectivity

Proof. **Goal:** We want to show that $\forall y \in S \cup T, \exists n \in C \ni h(n) = y$

Case 1 ($y \in S$):

Let $y \in S$. Since f is surjective, $\exists k \in C$ s.t. $f(k) = y$.

Let $n = 2k$. By definition n is even. Observe:

$$h(n) = f\left(\frac{2k}{2}\right) = f(k) = y$$

Case 2 ($y \in T$):

Let $y \in T$. Since g is surjective, $\exists k \in C \ni g(k) = y$

Let $n = 2k - 1$. Then n is odd. Observe:

$$h(n) = g\left(\frac{(2k-1)+1}{2}\right) = g\left(\frac{2k}{2}\right) = g(k) = y$$

As every element $y \in S \cup T \ni n \in C \ni h(n) = y \implies h : C \rightarrow S \cup T$ is surjective. □

Injectivity

Proof. **Goal:** We want to show that for arbitrary elements $n_1, n_2 \in C, h(n_1) = h(n_2) \implies n_1 = n_2$.
Suppose $h(n_1) = h(n_2) = y$.

Case 1 (Same Parity):

Suppose n_1, n_2 be both even. Then the following statements hold.

$$h(n_1) = h(n_2) \implies f\left(\frac{n_1}{2}\right) = f\left(\frac{n_2}{2}\right).$$

$$\text{Since } f \text{ is injective: } \frac{n_1}{2} = \frac{n_2}{2} \implies n_1 = n_2$$

Suppose n_1, n_2 be both odd. Then the following statements hold.

$$h(n_1) = h(n_2) \implies g\left(\frac{n_1+1}{2}\right) = g\left(\frac{n_2+1}{2}\right)$$

$$\text{Since } g \text{ is injective: } \frac{n_1+1}{2} = \frac{n_2+1}{2} \implies n_1 = n_2$$

Case 2 (Mixed Parity):

WLOG suppose n_1 is even and n_2 is odd.

$$h(n_1) \in \text{Img}(f) = S$$

$$h(n_2) \in \text{Img}(g) = T$$

$$\text{Since } S \cap T = \emptyset \implies h(n_1) \neq h(n_2) \equiv f(n_1) \neq g(n_2)$$

As $n_1, n_2 \in C$, $h(n_1) = h(n_2) \implies n_1 = n_2$. The function $h(n) : C \rightarrow S \cup T$ is injective □

Conclusion

Since h is both surjective and injective, it is bijective.

As there exists a bijective function $h : C \rightarrow S \cup T \implies S \cup T$ is countable. □

Question 2: Sec 1.3 (q1)

Question: Prove that a nonempty set T_1 is finite iff $\exists$ a bijection from T_1 onto a finite set T_2 .

Case 1 $\rightarrow$:

Note: To prove this by "Pigeon Hole Principle" we must suppose $|T_1| = |T_2|$ else there does not exist a way to construct a bijection.

Proof. (**Show:** If T_1 is finite. Then $\exists$ a $f : T_1 \rightarrow T_2$, where T_2 is finite and f is bijective)

Suppose T_1 is finite. Consider a function $f : T_1 \rightarrow T_2$ defined as:

$$f := \{(a_i, b_j) | a_i \in T_1 \text{ and } b_j \in T_2\}$$

Let a_i be an arbitrary element defined to be (where all elements are distinct):

$$a_i = a_1, a_2, a_3, \dots, a_{|T_1|},$$

Let b_i be an arbitrary element defined to be (where all elements are distinct):

$$b_i = b_1, b_2, b_3, \dots, b_{|T_2|},$$

Clearly:

$$\forall b \in T_2, \exists a \in T_1 \ni f(a) = b$$

Thus f is surjective.

Clearly:

$$f(x) = f(y) \implies x = y$$

Thus f is injective

Conclusion: As f is both injective and surjective, given T_1 is finite then clearly there exists a bijection from T_1 onto T_2 . □

Case 2 $\leftarrow$:

Proof. (**Show:** If $\exists$ a $f : T_1 \rightarrow T_2$, where T_2 is finite and f is bijective. Then T_1 is finite.)

Suppose $\exists$ a $f : T_1 \rightarrow T_2$, where T_2 is finite and f is bijective.

For the sake of contradiction suppose T_1 is infinite.

As f is bijective we can construct an arbitrary mapping as:

$$f := \{(a_i, b_j) | a_i \in T_1 \text{ and } b_j \in T_2\}$$

Let a_i be an arbitrary element defined to be (where all elements are distinct):

$$a_i = a_1, a_2, a_3, \dots, a_{|T_1|},$$

Let b_i be an arbitrary element defined to be (where all elements are distinct):

$$b_i = b_1, b_2, b_3, \dots, b_{|T_2|},$$

However as T_1 is infinite and T_2 is finite then clearly $|T_2| < |T_1|$.

Thus for the function $f : T_1 \rightarrow T_2$ to be a function the following must be true (by "Pigeon Hole Principle"):

$$f(x) = f(y) \implies x \neq y$$

This is a contradiction, thus we can say: if $\exists$ a $f : T_1 \rightarrow T_2$, where T_2 is finite and f is bijective. Then T_1 is finite. $\square$

Question 3: Sec 1.3 (q4)

Proof. **Question:** Exhibit a bijection between $\mathbb{N}$ and the set of all odd integers greater than 13.

Setup

Let A be a set defined as:

$$S := \{z | z \in \mathbb{Z}, z > 13\}$$

Observe the elements of S , where $i \in \mathbb{N}$:

$$S_1 = 14$$

$$S_2 = 15$$

$$S_3 = 16$$

$$\dots$$

$$S_i = i$$

(Note: all elements defined are distinct)

Also let's define elements of $\mathbb{N}$ as follows, $j \in \mathbb{N}$:

$$n_1 = 1$$

$$n_2 = 2$$

$$n_3 = 3$$

$$\dots$$

$$n_j = j$$

(Note: all elements defined are distinct)

Lets construct a function that maps $\mathbb{N}$ to S :

$$f : \mathbb{N} \rightarrow S = \{(s_i, n_j) | s_i \in S \wedge n_j \in \mathbb{N}\}$$

We can also say this is the same as:

$$f(n) = 13 + n$$

Surjectivity

Proof. Goal: Show that $\forall s \in S, \exists n \in \mathbb{N} \ni f(n) = s$

Let $s \in S, n \in \mathbb{N}$:

$$s = 13 + n = f(n)$$

By algebra:

$$s = 13 + n$$

$$n = s - 13$$

Clearly s has a infima of 14 as defined by the proposition and must be an integer.

$$\text{Therefore } s - 13 \in \mathbb{N}$$

Back solving:

$$\begin{aligned} f(s - 13) &= (s - 13) + 13 \\ &= s - 13 + 13 \\ &= s + 13 \end{aligned}$$

Observe $f(n) = s + 13$ which we obtained and is the expected result as the input to the prior solution and no terms escape the defined sets; thus the statments holds

We have show that $\forall s \in S, \exists n \in \mathbb{N} \ni f(n) = s$.

Therefore f is surjective.

□

Injectivity

Proof. Goal: Show that given $a, b \in \mathbb{N}$ and $f(a) = f(b) \implies a = b$ Let $a, b \in \mathbb{N}$ and $f(a) = f(b)$.

Observe:

$$\begin{aligned} f(a) &= f(b) \\ a + 13 &= b + 13 \\ a + 13 - 13 &= b + 13 - 13 \\ a &= b \end{aligned}$$

Thus we have shown that given $a, b \in \mathbb{N}$ and $f(a) = f(b) \implies a = b$

Therefore f is injective.

□

Conclusion

As f is both injective and surjective we have shown there exists a bijection between $\mathbb{N}$ and the set of all odd integers greater than 13. $\square$

Question 4: Sec 1.3 (q6)

Question: Exhibit a bijection between $\mathbb{N}$ and a proper subset of itself.
Let's construct a set E , a set of all even natural numbers, and let $k \in \mathbb{N}$:

$$E := \{n | n = 2k\}$$

Clearly $1 \in \mathbb{N}$ and $1 \notin E$ as 1 is not even. Therefore $E \subset \mathbb{N}$.

To prove a bijection exists, we must construct a function $f : \mathbb{N} \rightarrow E$ that is surjective and injective. Let's construct the function as follows, let $n \in \mathbb{N}, v \in E$:

$$f(n) = 2n = v$$

Surjectivity

Proof. Goal: Show that $\forall s \in S, \exists n \in \mathbb{N} \ni f(n) = s$
Let $v \in E, n \in \mathbb{N}$. We can say the following:

$$v = 2n \implies \frac{v}{2} = n$$

Recall that $v \in E$, which constricts the elements of E to be even, as defined. Let $k \in \mathbb{N}$ and $v = 2k$. Thus we can rearrange the equation as follows:

$$\frac{v}{2} = n \implies \frac{2k}{2} = n \implies k = n$$

Now notice we have not violated any closure properties thus the solution remains valid.

Backsolve: to confirm validity.

$$f(k) = 2k = v$$

Recall: we had defined $v = 2k$, so this is the expected answer.

Conclusion:

Thus, we have show that $\forall s \in S, \exists n \in \mathbb{N} \ni f(n) = s$
Therefore f is surjective.

$\square$

Injectivity

Proof. Goal: Show that given $a, b \in \mathbb{N}$ and $f(a) = f(b) \implies a = b$

Let $a, b \in \mathbb{N}$ and $f(a) = f(b) \implies a = b$. Observe:

$$f(a) = f(b)$$

$$2a = 2b$$

$$\frac{2a}{2} = \frac{2b}{2}$$

$$a = b$$

Thus we have shown that given $a, b \in \mathbb{N}$ and $f(a) = f(b) \implies a = b$

Therefore f is injective.

□

Conclusion

As f is both injective and surjective we have shown there exists a bijection between $\mathbb{N}$ and the set of all odd integers greater than 13.

Question 5: Sec 1.3 (q7)

Question: Prove that a set T_1 is denumerable iff there is a bijection from T_1 onto a denumerable set T_2

Case 1 $\rightarrow$:

Prove: Given a denumerable set T_1 we can construct a bijection onto a denumerable set T_2 .

Proof. Suppose T_1 is denumerable. This implies a bijective function f exists:

$$f : \mathbb{N} \rightarrow T_1$$

Consider the existence of another denumerable set T_2 . Likewise a bijective function g must also exist:

$$g : \mathbb{N} \rightarrow T_2$$

As these functions are bijective, this implies the existence of an inverse function for each.

$$f^{-1} : T_1 \rightarrow \mathbb{N} \text{ and } g^{-1} : T_2 \rightarrow \mathbb{N}$$

Goal: we want to show there exists a function $h : T_1 \rightarrow T_2$ that is bijective.

Approach: consider the change in space

$$T_1 \rightarrow \mathbb{N} \rightarrow T_2$$

Consider the composition function:

$$g \circ f^{-1} : T_1 \rightarrow T_2$$

From definition above we know $Img(f^{-1}) = \mathbb{N}$, via the implication of bijection.

We also know that $g : \mathbb{N} \rightarrow T_2$ is bijective.

Thus we can say the following is a valid bijective function:

$$g : Img(f^{-1}) \rightarrow T_2 \equiv g : \mathbb{N} \rightarrow T_2$$

Conclusion: Thus we have shown given a denumerable set T_1 we can construct a bijection towards another denumerable set T_2 . □

Case 2 $\leftarrow$:

Prove: Given a denumerable set T_2 , if a bijection exists from T_1 onto T_2 then T_1 must be denumerable.

Proof. Suppose T_2 is a denumerable set. Thus, there must exist a bijection from T_2 and $\mathbb{N}$. Thus there must exist some functions:

$$f : \mathbb{N} \rightarrow T_2 \text{ and } f^{-1} : T_2 \rightarrow \mathbb{N}$$

Suppose there exists a bijection from T_1 onto T_2 . We can define the following functions:

$$g : T_1 \rightarrow T_2 \text{ and } g^{-1} : T_2 \rightarrow T_1$$

Approach: consider the change in space

$$T_1 \rightarrow T_2 \rightarrow \mathbb{N}$$

We can construct the composition:

$$f^{-1} \circ g : T_1 \rightarrow \mathbb{N}$$

Consider the image set:

$$Img(g) = T_2$$

By substitution:

$$f^{-1}(Img(g)) = \mathbb{N}$$

As these functions are bijective, we can consider their inverses.

$$f(\mathbb{N}) = T_2 \implies g(T_2) = T_1 \implies g(f(\mathbb{N})) = T_1$$

This means there exists a bijective function from $\mathbb{N} \rightarrow T_1$. Therefore, T_1 must be denumerable.

Conclusion: We have shown given a denumerable set T_2 , if a bijection exists from T_1 onto T_2 then T_1 must be denumerable. □

Question 6: Sec 1.3 (q8)

Question: Give an example of a countable collection of finite sets whose union is not finite.

Consider the sets defined to be:

$$A_n = \{n\}, \forall n \in \mathbb{N}$$

These sets are finite sets defined as:

$$A_1 = \{1\}$$

$$A_2 = \{2\}$$

$$A_3 = \{3\}$$

...

$$A_n = \{n\}$$

Observe each set $A_1, A_2, A_3, \dots, A_n$ are finite (singleton) sets.

Though the union of all sets, by how we defined $A_1, A_2, A_3, \dots, A_n$, would be result in the equivalent set $\mathbb{N}$ which is not finite.